

Nội dung

- 1 Problem introduction
- 2 Decision Tree for classification
- 3 Decision Tree for regression
- 4 Pruning Tree

Gender prediction

Phân tích dữ liệu thông minh

Decision Tree

TS. Nguyễn Tiến Huy
ntienhuy@fit.hcmus.edu.vn

#No	Age	Weight	Height	Gender
1	54	65	181	Male
2	19	82	173	Male
3	24	54	165	Female
4	54	57	170	Female
5	18	81	185	Male
6	51	45	155	Female

Gender prediction

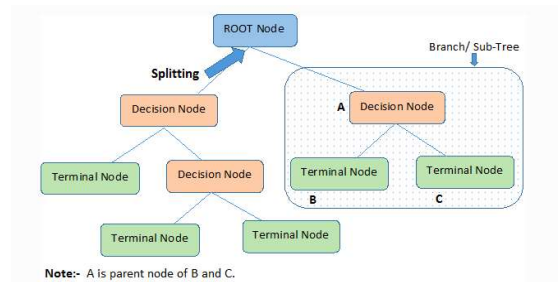
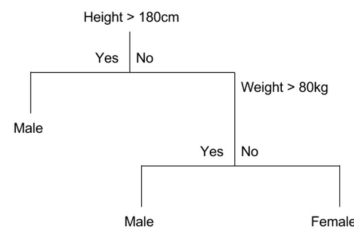
Decision Tree

How to build a decision tree

• CART (Classification and Regression Trees) → uses **Gini Index(Classification)** as metric.

• ID3 (Iterative Dichotomiser 3) → uses **Entropy function** and **Information gain** as metrics.

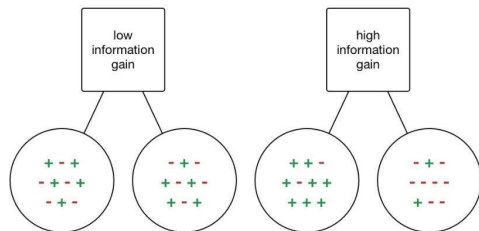
#No	Age	Weight	Height	Gender
1	54	65	181	Male
2	19	82	173	Male
3	24	54	165	Female
4	54	57	170	Female
5	18	81	185	Male
6	51	45	155	Female



Entropy and Information Gain

Weather prediction using Information Gain

Weather prediction



Day	Outlook	Temperature	Humidity	Wind	Play cricket
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Day	Outlook	Temperature	Humidity	Wind	Play cricket
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

$$H(S) = \sum_{c \in C} -p(c) \log_2 p(c)$$
$$H(S) = -\left(\frac{9}{14}\right) \log_2 \left(\frac{9}{14}\right) - \left(\frac{5}{14}\right) \log_2 \left(\frac{5}{14}\right)$$
$$H(S) = 0.94$$

Weather prediction

Day	Outlook	Temperature	Humidity	Wind	Play cricket
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Weather prediction

Day	Outlook	Temperature	Humidity	Wind	Play cricket
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Weather prediction

Day	Outlook	Temperature	Humidity	Wind	Play cricket
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

$$IG(S, Wind) = H(S) - \sum_{t \in T} P(t) * H(t)$$

$$P(S_{weak}) = \frac{\text{Number of weak}}{\text{Total}}$$

$$= \frac{8}{14}$$

$$P(S_{strong}) = \frac{\text{Number of strong}}{\text{Total}}$$

$$= \frac{6}{14}$$

$$IG(S, Wind) = H(S) - \sum_{t \in T} P(t) * H(t)$$

$$H(S_{weak}) = - \left(\frac{6}{8}\right) \log_2 \left(\frac{6}{8}\right) - \left(\frac{2}{8}\right) \log_2 \left(\frac{2}{8}\right)$$

$$= 0.811$$

$$H(S_{strong}) = - \left(\frac{3}{6}\right) \log_2 \left(\frac{3}{6}\right) - \left(\frac{3}{6}\right) \log_2 \left(\frac{3}{6}\right)$$

$$= 1$$

$$IG(S, Wind) = H(S) - \sum_{t \in T} P(t) * H(t)$$

$$= H(S) - P(S_{weak}) * H(S_{weak}) - P(S_{strong}) * H(S_{strong})$$

$$= 0.94 - \left(\frac{8}{14}\right) (0.811) - \left(\frac{6}{14}\right) (1.0)$$

$$= 0.048$$

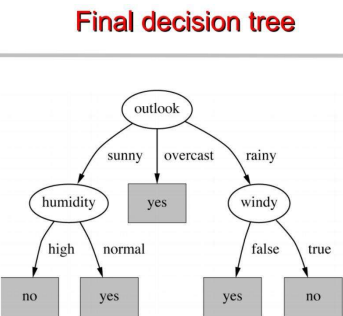
Weather prediction



Weather prediction



Weather prediction



Weather prediction using Gini index

Day	Outlook	Temperature	Humidity	Wind	Play cricket
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Weather prediction using Gini index

Day	Outlook	Temperature	Humidity	Wind	Play cricket
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

$$Gini(Outlook = Sunny) = 1 - P(Yes)^2 - P(No)^2$$

$$= 1 - (2/5)^2 - (3/5)^2$$

$$= 0.48$$

$$Gini(Outlook = Overcast) = 1 - (4/4)^2 - (0/4)^2$$

$$= 0$$

$$Gini(Outlook = Rain) = 1 - (3/5)^2 - (2/5)^2$$

$$= 0.48$$

$$Gini(S, Outlook) = (5/14) * 0.48 + (4/14) * 0 + (5/14) * 0.48$$

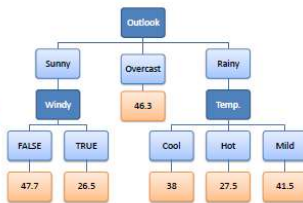
$$= 0.342$$

Categorical vs Numerical data

- **Numerical data** has meaning as a measurement, such as a person’s height, weight, IQ, or blood pressure
- **Categorical data** represents characteristics such as a person’s gender, marital status, hometown, or the types of movies they like.

Decision tree for regression

	Outlook	Temp	Humidity	Windy	Hours Played
0	Rainy	Hot	High	False	25
1	Rainy	Hot	High	True	30
2	Overcast	Hot	High	False	46
3	Sunny	Mild	High	False	45
4	Sunny	Cool	Normal	False	52
5	Sunny	Cool	Normal	True	23
6	Overcast	Cool	Normal	True	43
7	Rainy	Mild	High	False	35
8	Rainy	Cool	Normal	False	38
9	Sunny	Mild	Normal	False	46
10	Rainy	Mild	Normal	True	48
11	Overcast	Mild	High	True	52
12	Overcast	Hot	Normal	False	44
13	Sunny	Mild	High	True	30



Decision tree for regression

- Standard deviation

Hours Played
25
30
46
45
52
23
43
35
38
46
48
52
44
30

$$\text{Count} = n = 14$$

$$\text{Average} = \bar{x} = \frac{\sum x}{n} = 39.8$$

$$\text{Standard Deviation} = S = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} = 9.32$$

$$\text{Coefficient of Variation} = CV = \frac{S}{\bar{x}} * 100\% = 23\%$$

Decision tree for regression

- Standard deviation for a split

$$S(T, X) = \sum_{c \in X} P(c) S(c)$$

		Hours Played (StDev)	Count
Outlook	Overcast	3.49	4
	Rainy	7.78	5
	Sunny	10.87	5
			14



$$\begin{aligned} S(\text{Hours}, \text{Outlook}) &= P(\text{Sunny}) * S(\text{Sunny}) + P(\text{Overcast}) * S(\text{Overcast}) + P(\text{Rainy}) * S(\text{Rainy}) \\ &= (4/14) * 3.49 + (5/14) * 7.78 + (5/14) * 10.87 \\ &= 7.66 \end{aligned}$$

Decision tree for regression

		Hours Played (StDev)
Outlook	Overcast	3.49
	Rainy	7.78
	Sunny	10.87
		SDR=1.66

		Hours Played (StDev)
Humidity	High	9.36
	Normal	8.37
		SDR=0.28

		Hours Played (StDev)
Temp.	Cool	10.51
	Hot	8.95
	Mild	7.65
		SDR=0.17

		Hours Played (StDev)
Windy	False	7.87
	True	10.59
		SDR=0.29

$$SDR(T, X) = S(T) - S(T, X)$$

$$\begin{aligned} SDR(\text{Hours}, \text{Outlook}) &= S(\text{Hours}) - S(\text{Hours}, \text{Outlook}) \\ &= 9.32 - 7.66 = 1.66 \end{aligned}$$

Decision tree for regression

	Outlook	Temp	Humidity	Windy	Hours Played
Sunny	Sunny	Mild	High	FALSE	45
		Cool	Normal	FALSE	52
		Cool	Normal	TRUE	23
		Mild	Normal	FALSE	46
		Mild	High	TRUE	30
Overcast	Overcast	Hot	High	FALSE	46
		Cool	Normal	TRUE	43
		Mild	High	TRUE	52
		Hot	Normal	FALSE	44
Rainy	Rainy	Hot	High	FALSE	25
		Hot	High	TRUE	30
		Mild	High	FALSE	35
		Cool	Normal	FALSE	38
		Mild	Normal	TRUE	48

How to stop

- Coefficient of variant.
- The number of samples in a branch.

Pruning Tree

- **Pruning Tree: to avoid overfitting.**
- How:
 - Control tree's depth
 - Control the number of samples in each node.

Cảm ơn đã theo dõi!